

# Off-limb 3D modeling of Ca II K

## Prior state of the pipeline

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**Goal.** Produce as realistic as possible a model of the off-limb Ca II K emission (to interpret Sunrise/SUSI data), starting from a 3D MURaM cube plus the Ca II level populations of the line.

## Pipeline (three stages)

1. **Geometry preparation** — `extend_cube.ipynb`. Reads a 1.5D Lightweaver synthesis of a MURaM SSD/network snapshot (`...lwsynth.392.8.fits`  $\Rightarrow$  lower/upper level populations `pops[x,y,0,z]`) and the raw MURaM snapshot ( $T, P, n_e, v$ ). Cuts a shallow-angle ( $80^\circ$ ) line through the periodic box and *unrolls* it by modulo-folding into a long horizontal slab ( $\sim 300$  Mm, `n_steps`  $\approx 12500$ ), so a single box tiles into a full off-limb line of sight. Output: an HDF5 cube [`6, Nx, Nlos, Nz`] with  $T, P, n_e, v_{\text{LOS}}$  and the two level populations (upper/lower flipped in  $z$  for the  $\tau$  convention).
2. **3D solver** — `mpi_synt_cube_simple.py`. MPI overseer/worker, one task per  $x$ -slice. For each  $(x, z_{\text{tangent}})$  it builds a *curved ray* (`create_curved_grid`): a straight LOS in spherical geometry giving path  $s$ , position  $y$ , and geometric height  $z$  above the surface, with tangent height  $= z_{\text{index}} \cdot 20$  km. The slab is interpolated onto the ray; then `synth`  $\rightarrow$  `calc_op_em`  $\rightarrow$  `simple_formal_solution`.
3. **Physics** — `calc_op_em.py`. Ca II K line opacity/emissivity from the populations (Voigt profile, natural broadening only, Doppler width from  $T$ , shift from  $v_{\text{LOS}}$ ), plus LTE continuum (`contop.py`) with  $\eta_c = \chi_c B_\nu(T)$ .

## 1D reference (for calibration)

`fs_1d_los.py` / `generate_s_prd.py` solve FALC in NLTE *with PRD* (Lightweaver/Promweaver) and dump `disk_center_test.opem.f`  $\chi(z, \lambda)$  and  $\eta(z, \lambda)$  on 82 heights (0–2342 km)  $\times$  1001  $\lambda$ , on the *same* wavelength grid as the 3D run (392.8–394.8 nm). The line source function  $S = \eta/\chi$  there falls from  $\sim 1.4 \times 10^{-7}$  at the surface to  $\sim 1 \times 10^{-9}$  in the chromosphere (cgs intensity units).

## Current limitation: no real source function

In `synth()` the formal solution is bypassed:

$$\text{op, em} = \text{calc\_op\_em}(\dots); \quad \text{simple\_formal\_solution}(\text{op}, \underline{\text{op}}, ds) \Rightarrow S \equiv 1; \quad \text{return } 1 - e^{-\tau_\lambda}.$$

The emissivity is set equal to the opacity ( $S \equiv 1$ ), and even that result is discarded in favour of  $I_\lambda = 1 - e^{-\tau_\lambda}$  — a *pure-opacity* profile with a flat, unit source function.

**Consequences.** Morphology of optically-thin emission is acceptable ( $I \approx \tau_\lambda$ ), but the line is unphysical: the real Ca II K source function drops by  $\sim 2$  orders of magnitude from photosphere to chromosphere and saturates to 1 where the line is thick.  $K_1/K_2/K_3$  structure and absolute/relative intensities cannot be reproduced.

*Next step: replace  $S \equiv 1$  with a height-stratified source function taken from the 1D NLTE model — see [docs/02\\_source\\_function\\_implementation.pdf](#).*